

Interpreting Data

Solutions



Curriculum Ready

ACMSP: 248, 249, 250, 278, 283



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Page 5 questions

1. A group of people were asked how many languages they speak and this table was partly completed.

a Complete the table.

Number of languages (x)	Frequency (f)	fx	Cumulative frequency (cf)
1	20	$20 \times 1 = 20$	$0 + 20 = 20$
2	$38 - 20 = 18$	$18 \times 2 = 36$	$20 + 18 = 38$
3	$50 - 38 = 12$	$12 \times 3 = 36$	$38 + 12 = 50$
4	7	$4 \times 7 = 28$	$50 + 7 = 57$
5	3	$5 \times 3 = 15$	$57 + 3 = 60$
$\sum f = 60$		$\sum fx = 20 + 36 + 36 + 28 + 15 = 135$	

b Show that the mean is $\bar{x} = 2.25$.

The mean is the average score (number of languages) given by $\bar{x} = \frac{\sum fx}{\sum f}$.

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{135}{60} = 2.25$$

c What is the median?

The median is the the middle score when the scores are arranged in ascending order. Since there are 60 scores the median is the average of the two middle scores which are in positions 30 and 31.

$$\begin{aligned} \therefore \text{median} &= \frac{\text{score in } 30^{\text{th}} \text{ position} + \text{score in } 31^{\text{st}} \text{ position}}{2} \\ &= \frac{2 + 2}{2} \\ &= 2 \end{aligned}$$

d What is the mode?

The mode is the score with the highest frequency.

$$\therefore \text{mode} = 1$$

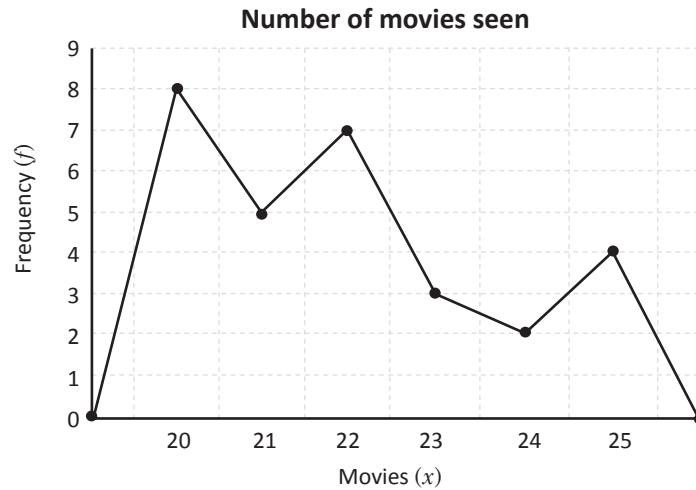
e Are $\sum f$ and the cumulative frequency of $x = 5$ equal? Why?

Yes, $\sum f$ and the cumulative frequency of $x = 5$ are equal as $x = 5$ is the highest score in the data.

This means all the scores are less than or equal to 5.

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2. A group of people were asked how many movies they had seen over the last year. The diagram below shows the frequency polygon for the results.



- a Use the diagram to complete the table below.

Movies (x)	Frequency (f)	fx	Cumulative frequency (cf)
20	8	$20 \times 8 = 160$	8
21	5	$21 \times 5 = 105$	$8 + 5 = 13$
22	7	$22 \times 7 = 154$	$13 + 7 = 20$
23	3	$23 \times 3 = 69$	$20 + 3 = 23$
24	2	$24 \times 2 = 48$	$23 + 2 = 25$
25	4	$25 \times 4 = 100$	$25 + 4 = 29$
$\sum f = 29$		$\sum fx = 636$	

- b What is the mean (to 2 decimal places if necessary)?

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{636}{29} = 21.93 \text{ (2 d.p.)}$$

- c What is the median?

There are 29 values in the data set. The median is the middle number and so is the 15th score.

$\therefore$ median = score in 15th position

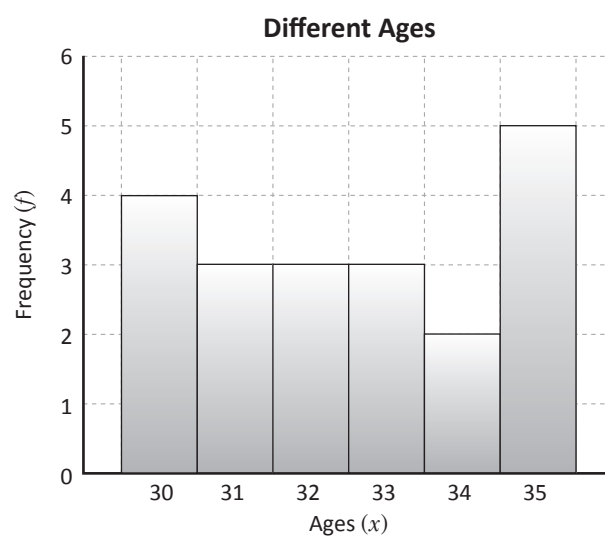
$$= 22$$

- d What is the mode?

The mode is the score with the highest frequency $\therefore$ mode = 20

Page 7 questions

3. A group of people were asked their age and this frequency histogram was produced.

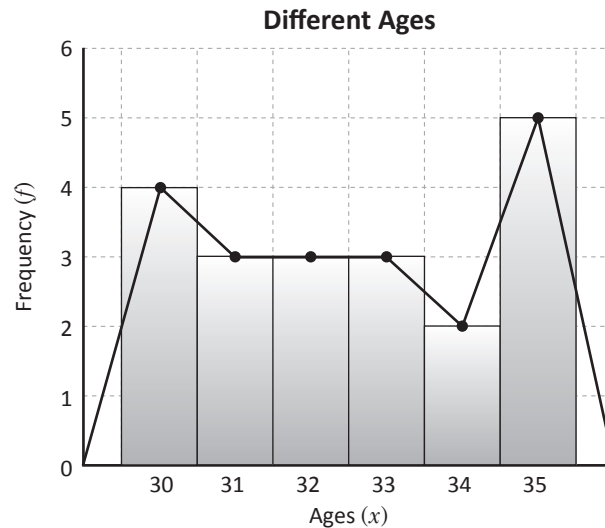


a Complete the table.

Ages (x)	Frequency (f)	fx	Cumulative frequency (cf)
30	4	$30 \times 4 = 120$	4
31	3	$31 \times 3 = 93$	$4 + 3 = 7$
32	3	$32 \times 3 = 96$	$7 + 3 = 10$
33	3	$33 \times 3 = 99$	$10 + 3 = 13$
34	2	$34 \times 2 = 68$	$13 + 2 = 15$
35	5	$35 \times 5 = 175$	$15 + 5 = 20$
$\sum f = 20$		$\sum fx = 651$	

Page 7 questions

- b Complete the polygon on the diagram.



- c Find the mean age (to 2 d.p. if necessary).

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$\bar{x} = \frac{651}{20}$$

$$= 32.55$$

- d What is the mode?

The mode is the score with the highest frequency.

$$\therefore \text{mode} = 35$$

- e What is the median?

There are 20 values in the data set. The median is the average of the values in 10th and 11th position in the data.

$$\therefore \text{median} = \frac{\text{score in 10}^{\text{th}} \text{ position} + \text{score in 11}^{\text{th}} \text{ position}}{2}$$

$$= \frac{32 + 33}{2}$$

$$= 32.5$$

Page 10 questions

1. Here is a list of data: 5, 7, 2, 3, 8, 0, 1, 2, 4, 8, 4, 2, 7, 8, 0, 1.

- a Arrange this data into ascending order.

$$0, 0, 1, 1 \mid \overset{Q_1}{2}, \overset{Q_2}{2}, \overset{Q_3}{2}, 3 \mid 4, 4, 5, 7 \mid 7, 8, 8, 8$$

There are 16 scores in total.

- b Find the median of this data set.

There are 16 scores. $\therefore$ the median is the average of the scores in 8th and 9th position.

$$\begin{aligned}\therefore \text{median} &= \frac{8^{\text{th}} \text{ score} + 9^{\text{th}} \text{ score}}{2} \\ &= \frac{3 + 4}{2} \\ &= 3.5\end{aligned}$$

- c Find Q_1 .

Q_1 is the average of the 4th and 5th scores: $\therefore Q_1 = \frac{1 + 2}{2} = 1.5$

- d Find the upper quartile.

Q_3 is the average of the 12th and 13th scores: $\therefore Q_3 = \frac{7 + 7}{2} = 7$.

- e Write down a 5-point summary of this data set.

- Lowest score = 0
- $Q_1 = 1.5$ (lower quartile)
- $Q_2 = 3.5$ (the median)
- $Q_3 = 7$ (upper quartile)
- Highest score = 8

- f What is the range of this data set?

Range = highest score – lowest score

$$\begin{aligned}\text{Range} &= 8 - 0 \\ &= 8\end{aligned}$$

Page 10 questions

- g What is in the interquartile range of this data set?

$$\begin{aligned}
 IQR &= Q_3 - Q_1 \\
 &= 7 - 1.5 \\
 &= 5.5
 \end{aligned}$$

Page 11 questions

2. A group of students' exam results are displayed in the table below.

Results in % (x)	Number of students (f)	Cumulative frequency (cf)
10	5	5
20	3	$5 + 3 = 8$
30	1	$8 + 1 = 9$
40	8	$9 + 8 = 17$
50	3	$17 + 3 = 20$
60	9	$20 + 9 = 29$
70	6	$29 + 6 = 35$
80	3	$35 + 3 = 38$
90	8	$38 + 8 = 46$
100	6	$46 + 6 = 52$
$\sum f = 52$		

- a Complete the cumulative frequency column.
- b How many students were used to create the table?
- c In which position is the median? What is the median?

The total number of students used is $\sum f = 52$

The median is the average of the 26th and 27th score.

$$\begin{aligned}
 \therefore \text{median} &= \frac{\text{score in 26}^{\text{th}} \text{ position} + \text{score in 27}^{\text{th}} \text{ position}}{2} \\
 &= \frac{60 + 60}{2} \\
 &= 60
 \end{aligned}$$

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- d Find Q_1 and Q_3 .

Q_1 is the average of the 13th and 14th scores:

$$\therefore Q_1 = \frac{40 + 40}{2} = 40$$

Q_3 is the average of the 39th and 40th scores:

$$\therefore Q_3 = \frac{90 + 90}{2} = 90$$

- e Find the interquartile range.

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 90 - 40 \\ &= 50 \end{aligned}$$

- f Write a 5-point summary on this data.

- Lowest score = 10
- $Q_1 = 40$ (lower quartile)
- $Q_2 = 60$ (the median)
- $Q_3 = 90$ (upper quartile)
- Highest score = 100

Page 15 questions

1. An athlete runs the same race 16 times. This is how long it takes him (in seconds) to run each time:

14, 12, 18, 14, 16, 18, 19, 14, 16, 17, 15, 13, 20, 16, 14, 19

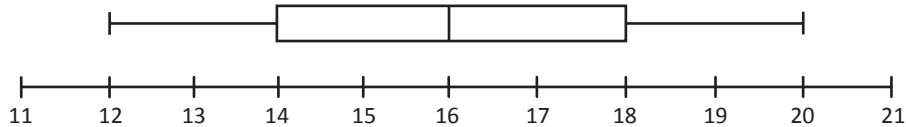
a Write these times in ascending order.

Q_1 Q_2 Q_3
 12, 13, 14, 14 | 14, 14, 15, 16 | 16, 16, 17, 18 | 18, 19, 19, 20

b Find a 5-point summary of this data.

- Lowest score = 12
- $Q_1 = \frac{14 + 14}{2} = 14$ (lower quartile) ← The average of the scores in the 4th and 5th positions
- $Q_2 = \frac{16 + 16}{2} = 16$ (the median) ← The average of the scores in the 8th and 9th positions
- $Q_3 = \frac{18 + 18}{2} = 18$ (upper quartile) ← The average of the scores in the 12th and 13th positions
- Highest score = 20

c Draw a box-and-whisker plot for this data.



d What is the range of the data?

$$\begin{aligned}
 \text{Range} &= \text{Highest score} - \text{lowest score} \\
 &= 20 - 12 \\
 &= 8
 \end{aligned}$$

e What is the interquartile range of the data?

$$\begin{aligned}
 IQR &= Q_3 - Q_1 \\
 IQR &= 18 - 14 \\
 &= 4
 \end{aligned}$$

Page 16 questions

2. During 8 days, a cat and a dog eat an amount of food (in grams) according to the table below.

	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Mon
Cat	70	100	40	90	50	70	55	100
Dog	65	100	90	80	75	85	50	85

a Arrange each set of data into ascending order.

Cat	40	50	55	70	70	90	100	100
Dog	50	65	75	80	85	85	90	100

Q_1 Q_2 Q_3

b Find a 5-point summary for each data set.

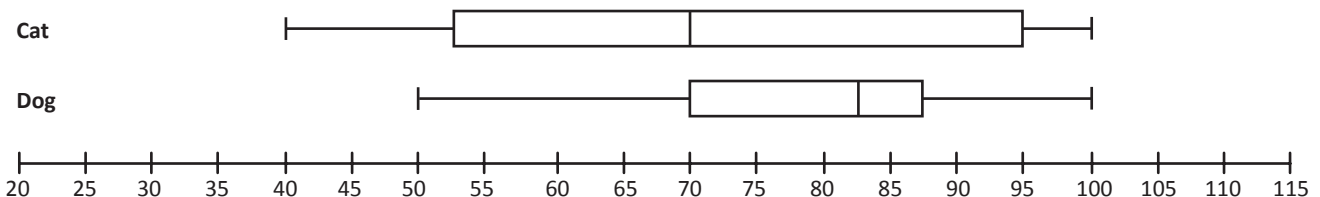
Cat:

- Lowest score = 40
- $Q_1 = \frac{50 + 55}{2} = 52.5$ (lower quartile) ← The average of the scores in the 2nd and 3rd positions
- $Q_2 = \frac{70 + 70}{2} = 70$ (the median) ← The average of the scores in the 4th and 5th positions
- $Q_3 = \frac{90 + 100}{2} = 95$ (upper quartile) ← The average of the scores in the 6th and 7th positions
- Highest score = 100

Dog:

- Lowest score = 50
- $Q_1 = \frac{65 + 75}{2} = 70$ (lower quartile) ← The average of the scores in the 2nd and 3rd positions
- $Q_2 = \frac{80 + 85}{2} = 82.5$ (the median) ← The average of the scores in the 4th and 5th positions
- $Q_3 = \frac{85 + 90}{2} = 87.5$ (upper quartile) ← The average of the scores in the 6th and 7th positions
- Highest score = 100

c Draw box-and-whisker plots for the different data sets on the number line below.



Page 16 questions

- d From the box-and-whisker plot, which had the greater interquartile range? Find the interquartile range.

From the box-and-whisker plot, the box for the cat is longer so the data for the cat had the greater interquartile range:

$$IQR(\text{cat}) = Q_3 - Q_1$$

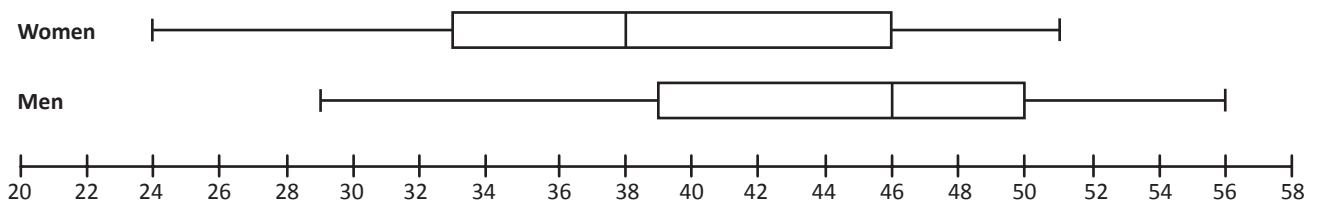
$$IQR(\text{cat}) = 95 - 52.5 = 42.5$$

$$IQR(\text{dog}) = Q_3 - Q_1$$

$$IQR(\text{dog}) = 87.5 - 70 = 17.5$$

Page 17 questions

3. 150 men and 150 women were in a survey and these are the resulting box-and-whisker plots from their ages:



- a How old was the youngest woman in the survey?

From the box-and-whisker plot the youngest woman was 24 years old.

- b How old was the oldest man in the survey?

The oldest man was 56 years old.

- c What is the median age for the women?

The median age for the women was 38 years old.

- d Find Q_1 for the men.

Q_1 for the men was 39 years old.

- e Find Q_3 for the women.

Q_3 for the women was 46 years old.

- f What is the age that half the men were older than?

The age that half the men were older than is the median. The median is 46 years old.

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4. Ava counted the number of books she read each month for a year. She wrote them in the table below:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
5	6	4	1	6	1	9	9	10	10	5	6

- a Find $\bar{x}$, the mean.

The mean number of books read each month $\bar{x} = \frac{\sum fx}{\sum f}$

$$\therefore \bar{x} = \frac{72}{12} = 6$$

- b Complete the table below for the above set of data.

x	$x - \bar{x}$	$(x - \bar{x})^2$
5	-1	$(-1)^2 = 1$
6	0	0
4	$4 - 6 = -2$	4
1	-5	25
6	$6 - 6 = 0$	0
1	-5	25
9	$9 - 6 = 3$	9
9	3	9
10	4	16
10	4	16
5	-1	1
6	0	0
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 106$

- c What is the formula to calculate σ_n ?

The formula for σ_n (standard deviation of a set of data containing n values) is:

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

where $\bar{x}$ is the mean

- d Show that the standard deviation of the above set of data to 2 decimal places is 2.97.

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{106}{12}} = \sqrt{8.833} = 2.97$$

Page 21 questions

1. Two movies received reviews from eight critics who gave the movie a score between 1 and 10. Here are their results:

Movie 1: 3, 8, 8, 6, 3, 5, 2, 5

Movie 2: 8, 7, 3, 4, 7, 10, 6, 3

- a Use this table to find the standard deviation of the Movies' scores:

First find the mean, then use the value of $\bar{x}$ to complete the tables:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Movie 1: $\bar{x} = \frac{40}{8} = 5$

Movie 2: $\bar{x} = \frac{48}{8} = 6$

Movie 1		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
3	$3 - 5 = -2$	4
8	$8 - 5 = 3$	9
8	$8 - 5 = 3$	9
6	$6 - 5 = 1$	1
3	$3 - 5 = -2$	4
5	$5 - 5 = 0$	0
2	$2 - 5 = -3$	9
5	$5 - 5 = 0$	0
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 36$

Movie 2		
Score (x)	$x - \bar{x}$	$(x - \bar{x})^2$
8	$8 - 6 = 2$	4
7	$7 - 6 = 1$	1
3	$3 - 6 = -3$	9
4	$4 - 6 = -2$	4
7	$7 - 6 = 1$	1
10	$10 - 6 = 4$	16
6	$6 - 6 = 0$	0
3	$3 - 6 = -3$	9
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 44$

Find the standard deviation based on the values of $\sum (x - \bar{x})^2$ in the tables.

Movie 1: $\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$
 $= \sqrt{\frac{36}{8}}$
 $= 2.12$

Movie 2: $\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$
 $= \sqrt{\frac{44}{8}}$
 $= 2.35$

Page 21 questions

- b Complete these tables for the movies:

First write the scores in order.

Movie 1: $2, 3 \mid 3, 5 \mid 5, 6 \mid 8, 8$

Movie 2: $3, 3 \mid 4, 6 \mid 7, 7 \mid 8, 10$

	Movie 1	Movie 2
Lowest	2	3
Q_1	$\frac{3+3}{2} = 3$	$\frac{3+4}{2} = 3.5$
Q_2	$\frac{5+5}{2} = 5$	$\frac{6+7}{2} = 6.5$
Q_3	$\frac{6+8}{2} = 7$	$\frac{7+8}{2} = 7.5$
Highest	8	10
σ_n	2.12	2.35

- c How are the scores for each movie skewed?

For Movie 1:

the mean = the median = 5

$\therefore$ the data is not skewed and the scores are distributed normally.

For Movie 2:

the mean = 6 and the median = 6.5,
median > mean

$\therefore$ the scores are skewed to the left.

- d Which movie has the more consistent scores?

	Movie 1	Movie 2
Range	$8 - 2 = 6$	$10 - 3 = 7$
Interquartile range	$7 - 3 = 4$	$7.5 - 3.5 = 4$
σ_n	2.12	2.35

Movie 1 has the more consistent scores than Movie 2. This is because the standard deviation of the scores for Movie 1 is less than for Movie 2.

Page 22 questions

2. At the olympics, divers receive a score between 1 and 10 each time they dive. These are the scores after 12 dives for the divers who came in first and second place.

Diver A

7, 8, 5, 7, 8, 6, 6, 5, 8, 5, 8, 5

Diver B

7, 6, 4, 5, 6, 5, 10, 9, 8, 6, 9, 9

- a Find a 5-point summary for each diver's scores.

Start by putting the scores for each diver in ascending order.

Diver A

$$5, 5, 5 \mid 5, 6, 6 \mid 7, 7, 8 \mid 8, 8, 8$$

Diver B

$$4, 5, 5 \mid 6, 6, 6 \mid 7, 8, 9 \mid 9, 9, 10$$

	Diver A	Diver B
Lowest	5	4
Q_1	$\frac{5+5}{2} = 5$	$\frac{5+6}{2} = 5.5$
Q_2	$\frac{6+7}{2} = 6.5$	$\frac{6+7}{2} = 6.5$
Q_3	$\frac{8+8}{2} = 8$	$\frac{9+9}{2} = 9$
Highest	8	10

- b Find the range and interquartile range of each diver's scores.

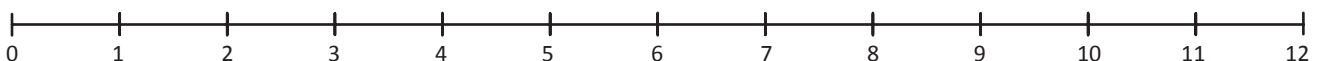
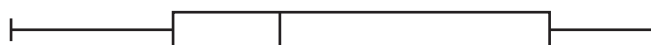
	Diver A	Diver B
Range	$8 - 5 = 3$	$10 - 4 = 6$
Interquartile range	$8 - 5 = 3$	$9 - 5.5 = 3.5$

- c Draw a box-and-whisker plot for each of the divers' scores. Are the scores skewed?

Diver A



Diver B



The scores for Diver A are not skewed. The scores for Diver B are skewed to the right (the box-and-whisker plot is longer on the right hand side of the median).

Page 23 questions

- d Draw up a table with the headings, **score (x)**, $x - \bar{x}$ and $(x - \bar{x})^2$ and use it to find σ_n for both divers.

Diver A

$$\bar{x} = \frac{78}{12} = 6.5$$

Diver A		
x	$x - \bar{x}$	$(x - \bar{x})^2$
5	-1.5	2.25
5	-1.5	2.25
5	-1.5	2.25
5	-1.5	2.25
6	-0.5	0.25
6	-0.5	0.25
7	0.5	0.25
7	0.5	0.25
8	1.5	2.25
8	1.5	2.25
8	1.5	2.25
8	1.5	2.25
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 19$

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\therefore \sigma_n = \sqrt{\frac{19}{12}}$$

$$= 1.258$$

Diver B

$$\bar{x} = \frac{84}{12} = 7$$

Diver B		
x	$x - \bar{x}$	$(x - \bar{x})^2$
4	-3	9
5	-2	4
5	-2	4
6	-1	1
6	-1	1
6	-1	1
7	0	0
8	1	1
9	2	4
9	2	4
9	2	4
10	3	9
$\sum (x - \bar{x}) = 0$		$\sum (x - \bar{x})^2 = 42$

$$\sigma_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\therefore \sigma_n = \sqrt{\frac{42}{12}}$$

$$= 1.87$$

- e If the winner is based on the total scores, which diver won?

The total score for Diver A was 78.

The total score for Diver B was 84.

$\therefore$ If the winner was based on total score, Diver B won.

- f Which diver had more consistent scores?

Diver A had the more consistent scores. This is shown by the lower standard deviation of Diver A's scores.

Page 24 questions**3. Answer these questions about skewness of data:**

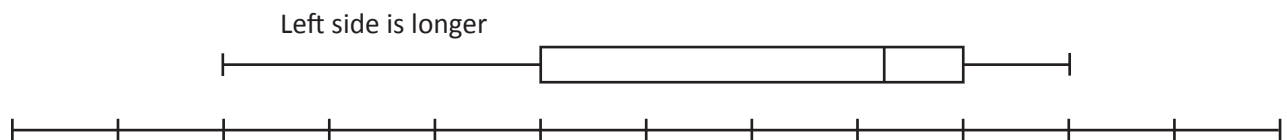
- a If the median is less than the mean which way would the data be skewed (according to the rule of thumb)?

The data is skewed to the right.

- b When is data skewed to the left (according to the rule of thumb)?

When median $>$ mean

- c Sketch a box-and-whisker plot representing data that is skewed to the left.



- d Standard deviation is a measure of how far each score is from the mean. What does this mean?

The standard deviation measures how spread out the scores are. The larger a standard deviation is, the further the scores are from the mean.

- e How is the standard deviation affected by the consistency of the scores?

The more consistent the scores, the lower the standard deviation. The less consistent the scores, the higher the standard deviation.

- f Does data with a higher or lower standard deviation have more consistency?

Data with a lower standard deviation has more consistency.



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